TRAFFIC AS A COMPRESSIBLE FLUID

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01 ABSTRACT AND INTRODUCTION

This project investigates road traffic as a one-dimensional compressible flow using the Lighthill-Whitham-Richards (LWR) model and its numerical implementation through the Cell Transmission Model (CTM). Although traffic consists of individual vehicles, many large-scale features of road motion can be described macroscopically using vehicle density, flow rate, and mean speed, in a similar way to a compressible fluid. This allows collective effects such as the build-up of congestion, the propagation of traffic jams, and the resulting changes in journey time to be studied without tracking each vehicle individually. The Greenshields model was used to provide a linear speed-density relation, allowing the formation and propagation of density waves and shock fronts to be examined. The CTM was then used to discretise the road into cells and evolve traffic density numerically in time. Using this approach, we investigated how traffic density develops under different conditions, how congestion emerges and propagates, and how this affects estimated travel times along the road.

02 THEORY

The LWR models traffic flow in a 1D description, similar to the analysis of an compressible fluid, dictated by mass conservation. Traffic is treated as having a macroscpic density ρ and flow rate q , governed by the core principles of conservation laws and the flow-density relation. The former can be derived directly from the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0$$

over a 1D road segment. Imposing vehicles to act as a mass and treating density as vehicles per unit length yields

$$\frac{\partial \rho}{\partial t} + \frac{\partial q}{\partial x} = 0$$

the conservation law disctating the LWR model. However, this equation contains unknowns, making it undetermined. A relationship used between q and ρ here is known as the fundamental diagram. Acting as as the experimentally obtained relation of traffic, this shows increasing flow with density to a critical point; after which congestion occurs causing reduced flow. The Greenshields model assumes a linear speed-density relation, yielding the fundamental diagram

$$q = \rho u = v_{max}\rho(1 - \frac{\rho}{\rho_{max}})$$

Applying this with the conservation condition gives a PDE describing characteristic wavespeed

$$\frac{\partial q}{\partial \rho} = v_{max}(1 - \frac{2\rho}{\rho_{max}})$$

highlighting the pertubations that propogate at speeds that depend on a local density. This forms shock waves that act as density discontinuities; defining the LWR model as a whole.

With the waves characteristics descibed by the LWR-Greenshields model, it becomes posible to determine how traffic density evolves with time. Prediciting the behaviour and estimaring travel time between two given points requires a numerical solution of the PDE shown above. The Cell Transition Model (CTM) provides a framework using discrete cells, meaning real traffic data can be applied to the LWR model, acting as a numerical solution to the characteristic wave behaviour.

03 METHODOLOGY

04 RESULTS

05 DISCUSSION

06 CONCLUSION

References: